

The Coordination Tax: Light-Speed Delay in a Self-Replicating Probe Swarm Bites Only When Hops Are Long

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Abstract—A swarm of self-replicating probes fills a stellar field star by star, each probe settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled, and names finite light-speed as its own future work. We add that light-speed limit: a probe deciding at a star treats a distant star as settled only once the news, travelling at the speed of light, has arrived, so probes race for the same target from stale views and waste trips. Over a 32-seed paired ensemble on identical galaxies, the delay slows the time to fill the field by a median of about 0 percent for powered nearest-neighbour flight, about 30 percent for nearest-star slingshot flight, and about 50 percent for maximum-boost slingshot flight; every case still reaches full coverage, so lag alone leaves no permanent unsettled fraction. The scale of the lag is set by a single dimensionless ratio, the information delay across a hop divided by the travel time across it, which equals the probe speed in units of the speed of light; whether it actually bites is decided by how non-local each hop is. Fast but local flight is nearly immune, because a probe that loses a race simply takes the star next door, while the cost falls on long-range hops made from stale views. Under finite light-speed the slingshot speed-up is real, but a meaningful fraction of it is eaten by uncoordinated long-range collisions.

Index Terms—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics

I. INTRODUCTION

A self-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself from local material, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. That a machine can build a copy of itself at all is von Neumann’s result, and a probe of this kind saturating the Galaxy in a few million years is the sharpest form of the Fermi question: if interstellar settlement is possible it should already be complete, so the observed silence is a paradox [1], [2]. How long the filling takes is the exploration-timescale question, and it has a long modelling lineage. Early work framed each interstellar probe as an independent agent acting on priors fixed before launch [3], and treated settlement as a front: a density-dependent diffusion with a travelling-wave solution [4], or a Monte Carlo migration filling the Galaxy in tens of millions of years [5]. Later simulations made the field concrete, sending probes and subprobes through a defined stellar volume to measure fill times directly [6], [7], and established the seeded Monte Carlo methodology that situates

an exploration timescale within a distribution of realizations rather than a single number [8].

The sharpest treatment for self-replicators is the slingshot line. Forgan, Papadogiannakis and Kitching showed that a single probe stealing energy from the motion of the stars it passes, a gravitational slingshot, explores up to two orders of magnitude faster than a probe under its own power [9]. Nicholson and Forgan extended that manoeuvre to self-replicating probes and confirmed the speed-up survives replication, with the surprising result that aiming for the nearest star can beat aiming for the largest boost [10]. The probe speeds that make this regime interesting, from the powered cruise of $3 \times 10^{-5} c$ up toward the 0.1 to 0.2 c of directed-energy or high-velocity concepts [11], [12], are exactly where the next assumption starts to hurt.

That assumption, which Nicholson and Forgan name as a limitation and leave for future work, is that every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another has just claimed, because it always knows. In a real galaxy no such oracle exists: news that a star has been settled can travel no faster than light, so a probe choosing its next target acts on a picture of the field that is, for distant stars, years or centuries out of date. This is the elementary consequence of a finite signal speed, that events have only a partial ordering and a remote event is unknowable until its signal arrives [13].

This paper adds that light-speed limit to the self-replicating slingshot model and measures what it costs. The question is not bookkeeping: it decides whether the celebrated slingshot speed-up survives contact with the finite speed of information, or whether a swarm racing outward on stale maps wastes much of the advantage colliding with itself. The answer, built from a controlled paired-ensemble experiment, is that the tax is real but sharply selective: its scale is set by the probe’s speed and its bite by how far each hop reaches. Fast-but-local flight escapes almost untouched; fast-and-far flight pays.

II. THE LIGHT-DELAYED BELIEF MODEL

A. The knowledge gate

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next

target at star f in year Y believes a distant star i is settled only once that beacon’s light has had time to arrive:

$$\text{settled}(i) + \frac{\text{dist}(f, i)}{c} \leq Y. \quad (1)$$

Under the perfect-information assumption of the source model, (1) collapses to “settled at any non-negative year,” and the model reproduces the earlier results bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival. This is availability chosen over consistency: rather than wait for a single agreed map that finite light-speed makes impossible, each probe acts now on its local view [13].

B. The three flight policies

We carry three named policies, following the slingshot studies we extend [9], [10]. *Powered nearest-neighbour* flight moves under the probe’s own power at a cruise speed of $3 \times 10^{-5} c$ to the nearest unsettled star. *Nearest-star slingshot* flight takes the same nearest target but gains speed from gravitational slingshots off the stars it passes. *Maximum-boost slingshot* flight instead selects the target that yields the largest velocity gain, deliberately reaching past near neighbours to distant stars. The three span the axis that matters here: powered flight is slow and local, nearest-star slingshot is fast and local, and maximum-boost slingshot is fast and far.

C. The lag ratio

How much the gate matters is governed by a single dimensionless ratio: the information delay across a hop divided by the travel time across the same hop. For a hop of length d the delay is d/c and the travel time is d/v , so the ratio is

$$\Lambda = \frac{d/c}{d/v} = \frac{v}{c}, \quad (2)$$

the probe’s speed in units of c , independent of the hop length. A slow probe ($\Lambda \ll 1$) outpaces its own ignorance only trivially: by the time it crosses a hop the news of that hop has long since propagated, so stale views barely arise. A fast probe carries a large Λ and routinely decides before the relevant news can reach it. At the powered cruise speed, $\Lambda \approx 3 \times 10^{-5}$ is negligible; the effect can only appear once probes are fast, which is precisely the slingshot regime.

D. Experimental design

The field is a box of stars at a uniform density of one star per cubic parsec, the convention of the source model; the real solar neighbourhood is sparser, near 0.14 stars per cubic parsec [14], which lengthens hops and thus the absolute timescales but not the relative penalties reported here. Two properties keep the measurement honest. First, knowledge is evaluated only at the decision star at the moment of decision; news a probe’s path happens to cross in mid-flight is ignored. This deliberately undercounts what a probe could know, so the measured waste is a conservative upper bound on the true penalty. Second, the

TABLE I
MEDIAN FILL-TIME PENALTY FROM LIGHT-SPEED LAG, OVER THE 32-SEED PAIRED ENSEMBLE AT 300 STARS. THE PENALTY TRACKS HOP LOCALITY, NOT RAW SPEED.

Flight policy	Median penalty (%)	IQR (%)
Powered nearest-neighbour	0.0	[0.0, 0.0]
Nearest-star slingshot	30.3	[20.0, 38.5]
Maximum-boost slingshot	51.4	[46.0, 54.0]

whole simulation is a pure, seeded, fixed-step fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That paired design is what lets us attribute a difference in fill-time to the delay alone rather than to the luck of the star layout.

III. RESULTS: THE COORDINATION TAX

We measured the cost as a paired ensemble: 32 seeded galaxies of 300 stars each, every one filled twice, once with perfect information and once with the light-speed gate of (1), so that the only difference between the two runs of a pair is the delay. The quantity of interest is the relative slowdown in the time to settle the entire field, t_{100} , per seed. Reporting the distribution across the ensemble rather than a single run follows the seeded Monte Carlo practice of the field [8].

The medians and their spread are given in Table I and shown per seed in Fig. 1. Powered flight pays essentially nothing; nearest-star slingshot loses about a third of its time; maximum-boost slingshot loses about half. Two facts stand out.

The first is that the penalty is *not* simply $\Lambda = v/c$ from (2). Powered flight is nearly untouched even though it is the slowest regime, and the two fast slingshot regimes differ from each other by nearly a factor of two despite comparable speeds. If Λ alone set the cost, the fastest probes would pay the most in a smooth progression; instead the cost tracks hop locality. A probe flying to the *nearest* star that loses a race has lost almost nothing: the next-nearest unsettled star is right beside it, so the wasted trip and the recovery are both short. A probe chasing the *biggest boost* deliberately reaches past its neighbours to a distant star; when it loses that race, both the wasted trip and the re-targeting detour are long. So $\Lambda \approx v/c$ sets the *scale* of the lag, but the *non-locality of each hop decides whether it bites*. Fig. 2 shows the effect on a single representative galaxy for the nearest-star policy: the light-speed run reaches full settlement at about 110,000 years against 80,000 for the perfect-information run on the same seed, a right-shift of roughly a third that is visible as a lag in the settlement curve, not a change in its final height.

The second fact is what does *not* happen: every case still fills the field to full coverage. Light-speed lag makes the swarm slower and more wasteful, but it never leaves a permanent unsettled remainder. Re-targeting guarantees that a probe which loses a race simply picks another star, and eventually the field is closed. A permanent settled fraction below one, a steady state in which some region stays forever

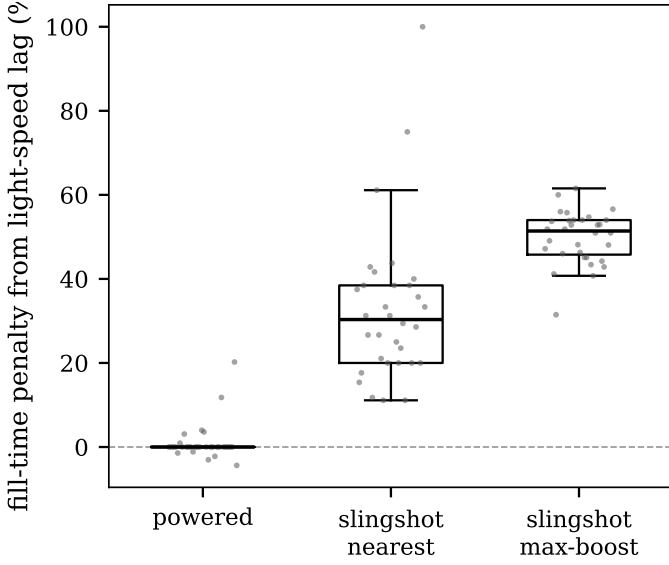


Fig. 1. Distribution of the per-seed fill-time penalty imposed by light-speed-limited coordination, over the fixed 32-seed ensemble at 300 stars, for the three flight policies (box: interquartile range and median; points: individual seeds). Powered nearest-neighbour flight pays essentially no penalty, because a lost race is recovered at the star next door, while the long-range maximum-boost policy fills the field materially later, because wasted trips launched from stale views become long detours.

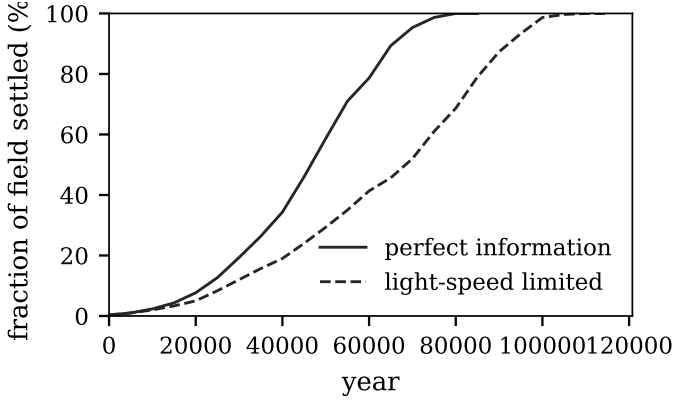


Fig. 2. Fraction of the reachable field settled versus year for the nearest-star slingshot policy on a single representative galaxy, under perfect global information (solid) and under light-speed-limited coordination (dashed). The light-speed curve is shifted later, reaching full settlement at about 110,000 years versus 80,000, a delay attributable to information lag alone; both curves reach the same final height.

empty, would require settled sites to *die*, adding a loss term to the dynamics; delay by itself supplies no such term. Lag is a tax on speed, not a barrier to completion.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture. Nicholson and Forgan’s speed-up is real and large [10], but it was computed under perfect coordination, and coordination is not free. Under finite light-speed a meaningful fraction of the advantage, roughly a third for nearest-star slingshots and half for maximum-boost, is spent on trips to stars that turn out to

be already taken. The more aggressively a policy reaches for distant, high-boost targets, the more of its speed it loses to uncoordinated collisions.

A. Limitations

The measurement is a conservative upper bound in two senses, and both point to future work. Probes here learn only at their decision stars, ignoring news their paths cross in flight, so a two-endpoint knowledge model would recover some of the wasted trips; and probes here share nothing directly, so adding probe-to-probe relay of the settled map would let a swarm partially coordinate rather than merely collide. Both would lower the tax without erasing it, since the fundamental limit, that no probe can know a distant star is taken until its light arrives, is set by physics, not by protocol. Two further simplifications bound the scope. The field is at a uniform one star per cubic parsec; a realistic, clumpier distribution would lengthen the long-range hops that carry the penalty, so the reported figures are conservative for the maximum-boost policy in particular. And the dynamics have no loss term: settled sites never fail, so the field always closes. A settlement-death term of the kind that produces the “Aurora” steady state, in which the settled fraction equilibrates below one [12], is exactly what would be needed to turn lag into a permanent unsettled remainder; delay alone does not.

B. The signature of delayed agreement

The behaviour is the expected signature of delayed multi-agent agreement. Formal consensus degrades as the ratio of communication delay to decision cadence grows: the standard protocol is stable only while the one-hop delay stays below a bound set by the coupling [15], a result later generalized to switching topologies and link failures [16]. Our probes never reach agreement about the settled map at all; at a mean inter-star spacing of about a parsec the round-trip light-time is several years, far longer than any decision cadence, so they behave like agents forced to act on purely local, out-of-date views. In the limit of unbounded delay, guaranteed agreement is not merely slow but impossible [17], and a system that must nonetheless keep acting is forced to trade a single consistent map for availability [18], which is precisely the choice (1) encodes.

This is the same progression that drove the space teleoperation and networking communities away from real-time control. Past about a second of round-trip delay a human operator abandons closed-loop control for move-and-wait [19], and as delay grows the handoff continues through supervisory control to full remote autonomy [20]. Past minutes to hours no continuous end-to-end path exists at all, and communication becomes hop-by-hop store-and-forward, the delay-tolerant regime motivated by the interplanetary case [21] and standardized first as an architecture [22] and now as a deployed protocol [23]. At interstellar hop lengths every one of those faster regimes is left far behind; each probe is effectively an independent colony acting on priors, and the coordination tax is what that independence costs when probes want the same stars.

C. A design lesson

There is a practical corollary. For a real swarm on stale maps, a locally greedy policy is not merely simpler than a boost-maximising one; it is more robust to the delay it cannot avoid, because a lost race costs only the star next door. A designer who cannot buy coordination should prefer locality, spending the slingshot advantage on speed rather than on reach.

V. CONCLUSION

Adding a finite speed of information to the self-replicating slingshot model turns a convenient assumption into a measurable cost. Over a paired ensemble of identical galaxies, light-speed lag slows the time to fill the field by a median of about 0 percent under powered flight, about 30 percent under nearest-star slingshots, and about 50 percent under maximum-boost slingshots, while every case still reaches full coverage. The scale of the tax is set by the dimensionless ratio $\Lambda = v/c$, but whether it bites is decided by how non-local each hop is: fast-but-local flight is nearly immune, and the cost falls almost entirely on the long-range hops that made the swarm fast in the first place. The slingshot speed-up survives contact with the finite speed of information, but a real swarm pays a coordination tax that a perfect-information model hides, and pays it in proportion to its reach. Two extensions would tighten the estimate without changing its shape: knowledge gained in flight, and probe-to-probe relay of the settled map. Both would lower the tax; neither can remove the physical floor beneath it.

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